

Лемма с прошлой лекции

$$1) v(x, t) \in C^2(\bar{Q}_{x^0, t^0})$$

$$2) v_{tt} - \Delta v = 0 \text{ в } Q_{x^0, t^0}$$

$$3) v|_{t=0} = 0, x \in \bar{Q}_{x^0, t^0}$$

$$4) v_t|_{t=0} = 0, x \in \bar{Q}_{x^0, t^0}$$

$$1), 2), 3), 4) \Rightarrow v(x, t) \equiv 0 \text{ в } \bar{Q}_{x^0, t^0}$$

Δ . (продолжение)

$$0 = \iiint_{Q_{\xi, \tau}} \operatorname{div} \bar{A}(\bar{x}) d\bar{x} = \iint_{\partial Q_{\xi, \tau}} (\bar{A}(\bar{x}), \bar{n}) dS = \iint_{\text{Б.П.}} (\bar{A}, \bar{n}) dS + \iint_{\text{Н.Кр.}} (\bar{A}, \bar{n}) dS = \iint_{\text{Б.П.}} (\bar{A}, \bar{n}) dS$$

на нижн. крышке $(\bar{A}, \bar{n}) = -\frac{1}{2}(v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^2 + v_{x_4}^2) = 0$

Б.П. подконуса: $\underbrace{(x_1 - \xi_1)^2 + (x_2 - \xi_2)^2 + (x_3 - \xi_3)^2 - (x_4 - \tau)^2}_{S(x_1, x_2, x_3, x_4)} = 0$

$$\bar{N} = \begin{pmatrix} x_1 - \xi_1 \\ x_2 - \xi_2 \\ x_3 - \xi_3 \\ -(x_4 - \tau) \end{pmatrix}$$

$$|\bar{N}|^2 = (x_1 - \xi_1)^2 + (x_2 - \xi_2)^2 + (x_3 - \xi_3)^2 + (x_4 - \tau)^2 = \{(x_1 - \xi_1)^2 + (x_2 - \xi_2)^2 + (x_3 - \xi_3)^2 - (x_4 - \tau)^2\} + 2(x_4 - \tau)^2 = 2(x_4 - \tau)^2$$

$$\Rightarrow |\bar{N}| = \sqrt{2} |x_4 - \tau|$$

на БП $\bar{n} = \begin{pmatrix} -\frac{x_1 - \xi_1}{\sqrt{2} |x_4 - \tau|} \\ -\frac{x_2 - \xi_2}{\sqrt{2} |x_4 - \tau|} \\ -\frac{x_3 - \xi_3}{\sqrt{2} |x_4 - \tau|} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

$$-\frac{x_4 - \tau}{\sqrt{2} |x_4 - \tau|} > 0 \Rightarrow |x_4 - \tau| = -(x_4 - \tau)$$



$$(\bar{A}, \bar{n}) = \frac{1}{2\sqrt{2}} \left[2v_{x_1} v_{x_4} \frac{x_1 - \xi_1}{(x_4 - \tau)} + 2v_{x_2} v_{x_4} \frac{x_2 - \xi_2}{(x_4 - \tau)} + 2v_{x_3} v_{x_4} \frac{x_3 - \xi_3}{(x_4 - \tau)} + v_{x_1}^2 + v_{x_2}^2 + v_{x_3}^2 + v_{x_4}^2 \right] =$$

$$\left\{ v_{x_1}^2 + 2v_{x_1} v_{x_4} \frac{x_1 - \xi_1}{x_4 - \tau} + \left(v_{x_4} \frac{x_1 - \xi_1}{x_4 - \tau} \right)^2 \right\} - \left(v_{x_4} \frac{x_1 - \xi_1}{x_4 - \tau} \right)^2$$

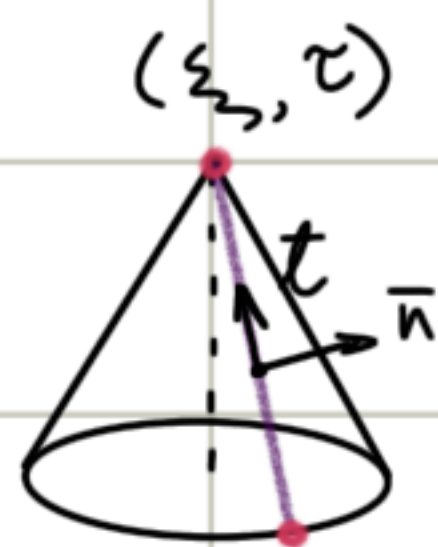
$$\begin{aligned} &= \frac{1}{2\sqrt{2}} \left[\left\{ v_{x_1} + v_{x_4} \frac{x_1 - \xi_1}{x_4 - \tau} \right\}^2 + \left\{ v_{x_2} + v_{x_4} \frac{x_2 - \xi_2}{x_4 - \tau} \right\}^2 + \left\{ v_{x_3} + v_{x_4} \frac{x_3 - \xi_3}{x_4 - \tau} \right\}^2 + v_{x_4}^2 \left(1 - \left(\frac{x_1 - \xi_1}{x_4 - \tau} \right)^2 - \left(\frac{x_2 - \xi_2}{x_4 - \tau} \right)^2 - \left(\frac{x_3 - \xi_3}{x_4 - \tau} \right)^2 \right) \right] \end{aligned}$$

$$\iint_{\text{Б.П.}} (\bar{A}, \bar{n}) dS = \iint_{\text{Б.П.}} \frac{1}{2\sqrt{2}} \left[\left\{ y^2 + \left\{ y^2 + \left\{ y^2 \right\} \right\} \right\} \right] dS = 0 \Leftrightarrow \begin{cases} y = 0 \\ y = 0 \\ y = 0 \end{cases}$$

$$\begin{cases} y = 0 \\ y = 0 \\ y = 0 \end{cases}$$

$$\begin{aligned} v_{x_1} &= -v_{x_4} \frac{x_1 - \xi_1}{x_4 - \tau} \\ v_{x_2} &= -v_{x_4} \frac{x_2 - \xi_2}{x_4 - \tau} \\ v_{x_3} &= -v_{x_4} \frac{x_3 - \xi_3}{x_4 - \tau} \end{aligned}$$

$$\begin{pmatrix} u_{x_1} \\ u_{x_2} \\ u_{x_3} \\ u_{x_4} \end{pmatrix} = u_{x_4} \begin{pmatrix} -\frac{x_1 - \xi_1}{x_4 - \tau} \\ -\frac{x_2 - \xi_2}{x_4 - \tau} \\ -\frac{x_3 - \xi_3}{x_4 - \tau} \\ 1 \end{pmatrix} = u_{x_4} \sqrt{2} \bar{n}$$



$$\frac{\partial u}{\partial t} = (\bar{\nabla} u, t) = (u_{x_4} \sqrt{2} \bar{n}, t) = u_{x_4} \sqrt{2} (\bar{n}, t) = 0$$

th

Задача

не может иметь более одного решения

$$\begin{cases} u_{tt} - \Delta u = F(x, t), & x \in \mathbb{R}^3, t > 0 \\ u|_{t=0} = \varphi(x), & x \in \mathbb{R}^3 \\ u_t|_{t=0} = \psi(x), & x \in \mathbb{R}^3 \end{cases}$$

где $\varphi(x), \psi(x), F(x, t)$ угод. необход. угод.

th

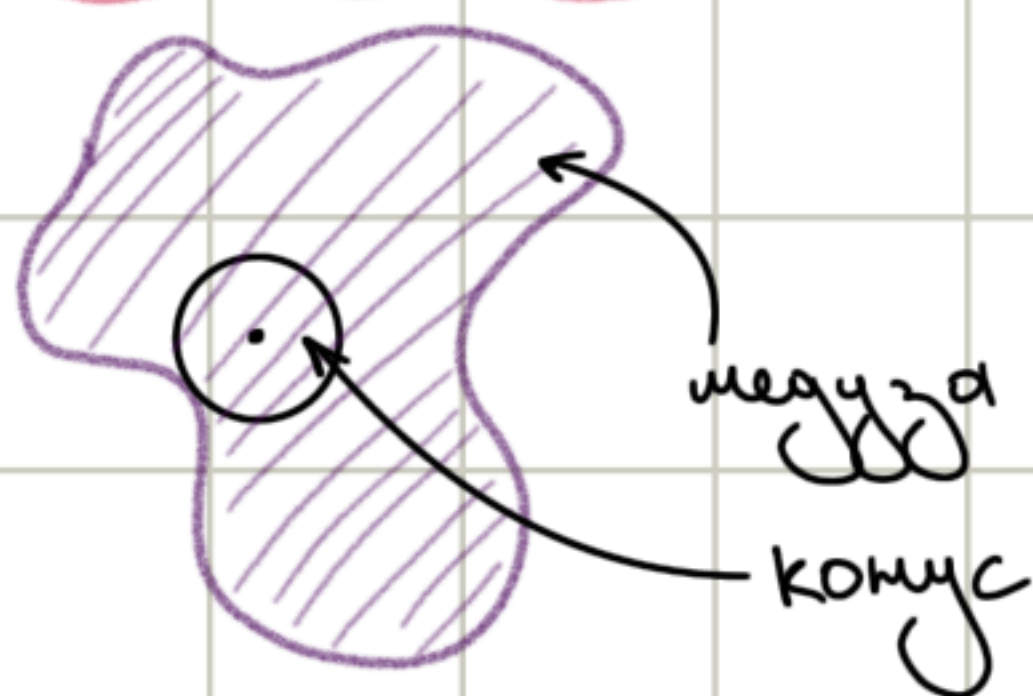
$$\nexists (1) u_{tt} - \Delta u = 0, x \in \mathbb{R}^3, t > 0$$

$$(2) u|_{t=0} = \varphi(x), x \in \mathbb{R}^3$$

$$(3) u_t|_{t=0} = \psi(x), x \in \mathbb{R}^3$$

$\exists \varphi(x) \in C^3(\mathbb{R}^3), \psi(x) \in C^2(\mathbb{R}^2)$, тогда реш. задачи (1), (2), (3) $\exists u$

$$u(x, t) = A\varphi(x, t) + \frac{\partial}{\partial t}(A\psi(x, t))$$



$u_1(x, t)$ - реш.
 $\in C(x \in \mathbb{R}^3, t > 0)$

$$\begin{cases} u_{tt} - \Delta u = F(x, t), & x \in \mathbb{R}^3, t > 0 \\ u|_{t=0} = \varphi_1(x) \in C^3(\mathbb{R}^3) \\ u_t|_{t=0} = \psi_1(x) \in C^2(\mathbb{R}^3) \end{cases}$$

$u_2(x, t)$ - реш.
 $\in C(x \in \mathbb{R}^3, t > 0)$

$$\begin{cases} u_{tt} - \Delta u = F(x, t), & x \in \mathbb{R}^3, t > 0 \\ u|_{t=0} = \varphi_2(x) \in C^3(\mathbb{R}^3) \\ u_t|_{t=0} = \psi_2(x) \in C^2(\mathbb{R}^3) \end{cases}$$

$$\nexists v(x, t) = u_1(x, t) - u_2(x, t)$$

реш.
 $v_{tt} - \Delta v = 0, x \in \mathbb{R}^3, t > 0$

$$v|_{t=0} = \varphi_1 - \varphi_2 = \Phi(x) \in C^3(\mathbb{R}^3)$$

$$v_t|_{t=0} = \psi_1 - \psi_2 = \Psi(x) \in C^2(\mathbb{R}^2)$$

$$v(x, t) = A_\Psi(x, t) + \frac{\partial}{\partial t}(A_\Phi(x, t)) = \frac{t}{4\pi} \iint_{|\eta|=1} \Psi(\bar{x} + \bar{\eta}t) dS_\eta + \frac{1}{4\pi} \iint_{|\eta|=1} \Phi(\bar{x} + \bar{\eta}t) dS_\eta + \frac{t}{4\pi} \iint_{|\eta|=1} (\bar{\nabla} \Phi, \bar{\eta}) dS_\eta$$

$$|v(x, t)| \leq \frac{t}{4\pi} \iint_{|\eta|=1} |\Psi(\bar{x} + \bar{\eta}t)| dS_\eta + \frac{1}{4\pi} \iint_{|\eta|=1} |\Phi(\bar{x} + \bar{\eta}t)| dS_\eta + \frac{t}{4\pi} \iint_{|\eta|=1} |(\bar{\nabla} \Phi, \bar{\eta})| dS_\eta$$

$\leq \|\bar{\nabla} \Phi\| |\eta| = \|\bar{\nabla} \Phi\|$

$$\forall \varepsilon > 0 \quad \forall T > 0 \quad \exists \delta = \delta(\varepsilon, T) > 0: \text{ если } |\varphi_1(x) - \varphi_2(x)| = |\Phi(x)| < \delta, \quad \forall x \in \mathbb{R}^3$$

$$|\psi_1(x) - \psi_2(x)| = |\Psi(x)| < \delta, \quad \forall x \in \mathbb{R}^3$$

$$\bar{\nabla} \varphi_1 - \bar{\nabla} \varphi_2 = \bar{\nabla}(\varphi_1 - \varphi_2) \Rightarrow \|\bar{\nabla} \Phi\| = \sqrt{(\varphi_1 - \varphi_2)'_{x_1}{}^2 + \dots} \quad \|\bar{\nabla}(\varphi_1 - \varphi_2)\| = \|\bar{\nabla} \Phi\| < \delta, \quad \forall x \in \mathbb{R}^3$$

$$\Rightarrow |u_1(x, t) - u_2(x, t)| < \varepsilon \quad \forall x \in \mathbb{R}^3, \quad \forall t: 0 \leq t \leq T$$

$$|v(x, t)| \leq \frac{T}{4\pi} \iint_{|\eta|=1} \delta dS_\eta + \frac{1}{4\pi} \iint_{|\eta|=1} \delta dS_\eta + \frac{T}{4\pi} \iint_{|\eta|=1} \delta dS_\eta = T\delta + \delta + T\delta = \delta(1+2T)$$

$$|v| \leq \varepsilon$$

$$\varepsilon = \delta(1+2T), \quad \delta = \frac{\varepsilon}{1+2T}, \quad \text{можно выбрать } \delta \in (0, \frac{\varepsilon}{1+2T}]$$

$$u(x_1, x_2, t)$$

$$u(x_1, x_2, x_3, t)$$